

计算机网络练习题(判断专题)

III、Decide true or false, give a brief explanation for each one.

- 1) Assume the propagation delay in a broadcast network is $3\mu\text{s}$ and the frame transmission time is $5\mu\text{s}$. The collision can be detected no matter where it occurs.
- 2) A IP address like 255.255.255.12 can be a mask in CIDR.
- 3) Rather than counting packets, TCP sequence numbers count bytes in the byte stream .
- 4) One role of a NIC is to accept a datagram from the next higher layer and encapsulate the datagram into a frame, along with frame header and trailer bit fields, such as rdt and CRC.
- 5) In packet-switched networks, the resources (buffers, link transmission rate) needed along a path to provide for communication between the end systems are reserved for the duration of the communication session between the end systems.

答案:

1. F。发送方需要在帧的最后一个比特发送前检测到碰撞。如果碰撞发生在接近目的地的位置,碰撞消息需要 $2\times 3=6\mu\text{s}$ 才能到达发送方。此时发送方已发送完整帧,不再监听碰撞。
2. F。我们首先将每个潜在掩码转换为二进制表示,然后检查其是否从左起连续出现多个1后接0。255.255.255.12对应的二进制为11111111 11111111 11111111 00001100。因此该二进制不能构成有效掩码。
3. T
4. T
5. F 不需要reservation